

The Theory of Effective Agents

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Introduction

The Theory of Efficient Agents (TEA) is a model for describing market dynamics in which pricing converges toward an efficient price (see the Efficient Market Hypothesis) through the interaction of efficient agents.

Foundations

The main focus of TEA is efficient agents—socio-economic phenomena that arise in highly liquid markets with minimal control by specific individuals, which is why they are more pronounced in commodity and cryptocurrency markets as well as in indices, and to a lesser extent in stock prices.

There are two types of agents, which are also divided into two parts based on how they manifest.

The reversal agent is the primary agent in TEA, whose function is to reverse the trend of price movement.

It is subdivided into a forerunner and an active phase.

The foreshadowing part signals an imminent reversal and also preserves the momentum of the previous active part, creating a hysteresis effect.

The active part exerts an influence that triggers the reversal. (This is a temporary assumption; it will be clarified later.) It also transmits the direction of action from the predictive part to the next reversal agent.

A discharge agent is an agent that arises when the driving part of the previous reversal agent has little or no influence.

It is subdivided into accumulative and active parts.

The accumulative part begins as the forerunner part of the reversal agent but then transitions into an accumulation phase due to a phenomenon called *a pullback*.

The pullback occurs rather chaotically, which creates market uncertainty. This phenomenon allows the price to move in the opposite direction while accumulating strength for a powerful impulse; the retracement lasts from the middle of the accumulation phase to the middle of the driving phase, to which it transfers all the accumulated strength, which then acts as the driving phase of the reversal agent.

Discharge agents have an accumulation effect, which allows them to line up one after another, building up momentum and releasing it in a single, amplified impulse. This phenomenon is called *an “discharge chain.”*

Technical Analysis of Agents

TEA was originally created to explain the values generated when using my personal EMAD indicator, which works by finding the intersection point of the slow SMA and the fast HMA using a modified secant method.

EMAD generates structures with which agent types are identified.

Reversal agents manifest as cubic parabolic patterns starting from $+\infty$, crossing 0, and then tending toward $-\infty$.

Positive EMAD values indicate the anticipatory phase, while negative values indicate the active phase.

Discharge agents appear as two classic parabolas; a positive one indicates the discharge phase, while a negative one indicates the active phase.

The manifestation of a *pullback* on the EMAD chart is intriguing. The process resembles a reverse retracement: values mirror each other, creating symmetrical patterns, and the price begins to move in the opposite direction until the process reaches the midpoint of the previous active phase. After that, the EMAD resumes its usual direction, and the active phase exerts a stronger influence on the price.

The Quasi-Probabilistic Nature of Agents

Initially, when studying the behavior of EMAD, there were attempts to interpret its values as trend strength, which turned out to be a crude approximation due to the presence of infinite values; for this reason, the application of classical probabilities also proved unsuccessful.

The most successful method for interpreting EMAD values has been the use of quasi-probabilities.

Positive EMAD values indicate a negative probability of a reversal, while negative values indicate a positive probability.

An example of negative probabilities in the context of EMAD in a dice game:

Let's imagine a game of dice, and say that rolling a 5 is negative, while rolling a 6 is zero. A 6 will never come up in this context. But the probability of rolling a 5 is not zero, which means it is possible, but in this case, the die will be flipped by an external factor—my hand.

This is often noticeable when working with EMAD; some spikes may occur prematurely, but if the EMAD value is positive and high enough (i.e., high negative probability), this fluctuation will be offset. The opposite can also occur: if the EMAD value is low but still negative, a random fluctuation may not be offset and, in some cases, may even be amplified by the emerging positive probability (a negative EMAD value), this is a rare special case, mainly observed on high time frames and at global levels.

The Fractal Nature of Agents

Agents exist at all scales, generating fractals; it is precisely the fractal nature of agents that accounts for the existence of offloading agents. For this reason, EMAD calculation requires fairly long periods, which most likely indicates a connection to the Kotelnikov (Nyquist–Shannon) theorem.

In this case, discharge agents function as synchronization mechanisms: if a low-order reversal agent has already moved into the active phase, while a high-order reversal agent is still in the leading portion and exerts a force in the opposite direction; this effect likely gives rise to the *rebound* phenomenon, and the transition of the highest-order reversal agent into the active portion provides the very gain that *discharge chains* create.

Connection to Socionomics

Moving beyond the level of technical analysis, one can assume that EMAD probabilities do not indicate the likelihood of a reversal per se, but rather the extent to which the market will react to new information. If the receipt of information—for example, bad news—coincides with a positive probability of a downward reversal, it will have a much more significant impact; but if it coincides with a negative probability, the information will most likely have little effect. The same applies to an upward reversal.

Also of interest is the correlation between the zero crossings, asymptotes, and extrema of the EMAD and the critical points of waves from Elliott Wave Theory—which is intriguing given the fractal nature of both EWT and TEA. Furthermore, this connection helps resolve the fundamental problem of subjectivity in wave counting.

Effectiveness and Vulnerabilities of TEA

It is most effectively applied in conjunction with Elliott Wave Theory, for the reasons outlined in the previous chapter. TEA also excellently illustrates global trends in cyclical markets.

However, due to the nature of EMAD and TEA itself, agents are not clearly visible in markets with a steady trend, which is why it is necessary to use logarithmic scales and price-derived instruments. A good example is the S&P 500 quotes, where it is difficult to discern the true behavior of agents—a problem that is corrected by using the Shiller Index.

The most prominent examples

- Brent (1999–2022). On the 2-week timeframe, the adverse influence of a reversal agent was evident from 1999 to 2001, triggering a discharge chain consisting of four discharge

agents that lasted from 2001 to May 2012, The forerunner phase of the subsequent upward reversal agent lasted through February 2015, when the zero line was crossed.

Subsequently, following the 2012–2016 reversal agent, a new downtrend chain consisting of two agents formed, ending with a zero-line crossover in mid-January 2022.

- Brent (2022–2025) Also on the 2-week timeframe, an upward reversal agent formed with a long lead-in phase; the zero crossover occurred in September 2025, and in February 2026, Operation “Epic Fury” began, leading to a surge in oil prices.
- Brent (2026–Present) On the 2-week timeframe, a zero crossover occurred in May, leading to a drop in oil prices; meanwhile, on the 3-week timeframe, this pattern began to transform into a discharge pattern, which could lead to a strong upward price momentum throughout 2026–2027.
- CAPE/S&P 500 (1982–2007) On the 1-month timeframe, a bearish reversal agent formed in 1982; a zero crossover occurred in 1987; and then, in 1990, a sell-off chain began to form. This pattern ended in February 2004 with a zero-crossing by a bullish reversal signal; then, in 2005, a bearish reversal signal began to form, the zero-crossing occurred in February 2007, and the driving phase lasted until February 2008, passing momentum to the next bullish reversal signal.
- CAPE/S&P 500 (2010–Present) Also on the 1-month timeframe, the forerunner phase of a bearish reversal signal began in 2010; the zero crossover occurred in August 2016; a correction phase began in March 2020; as of July 2026, we are in the middle of the active phase of the second discharge agent, with the peak of activity expected in late 2026–early 2027, which could lead to a major crisis. Interestingly, on the 3-month timeframe, we are in the active phase of the first discharge agent, which began in March 2002, following a downward reversal agent that began in 1984 and crossed zero in 1991.
- SI, ruble futures against the dollar. (2025–Present). On the weekly chart, an upward reversal pattern has been in progress since February 2025; it crossed zero in February 2026, and as of July 2026, we are at the peak of the active phase of this reversal pattern. On the 2-week timeframe, this reversal pattern has not yet crossed zero; if it does cross zero, the ruble could face a significant devaluation.